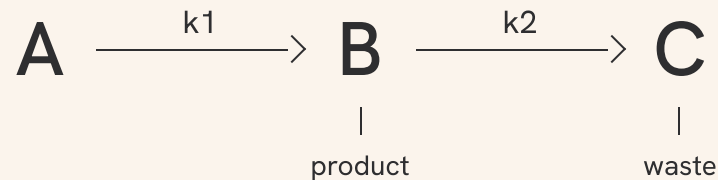




A reactor where the product *destroys* itself



B is the product and the feedstock for the waste reaction. Every second B spends in the reactor, some of it becomes C. Run too short and A never converts; run too long and B is destroyed.

What we predict

Yield of B from five operating knobs: feed rate, concentration, inlet temperature, reactor length, jacket temperature.

Why it matters industrially

The rigorous simulation is, in the problem statement's own words, "computationally expensive and too slow for real-time plant optimization." An engineer changing a setpoint cannot wait for a boundary-value solve.

150 operating conditions	Time
Our surrogate	24.8 ms
SciPy BDF solve, same system	5573 ms
Speed-up	225×

And the two agree to 0.017 yield-points.



Two activation energies, and their *ratio*

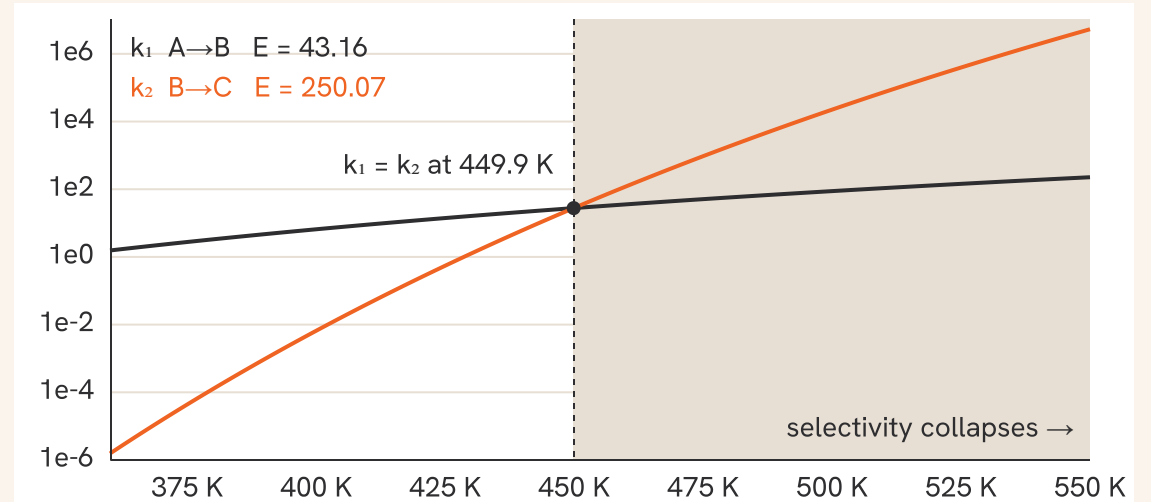
$$E_2/E_1 = 5.79$$

Heat accelerates destruction $\sim 6\times$ more steeply than formation. The rate constants cross at 449.9 K, inside the observed 351.6–548.0 K.

But it is not a temperature ceiling

Yield depends on $k \cdot \tau$, not k . Run hot with a short residence time and B leaves before it degrades: **49 training rows sit above the crossover and one reaches 75% yield**. The rule is joint in (T, τ) : above 449.9 K, every extra unit of residence time costs selectivity.

Opposite signs, near-equal magnitude: they cancel. Not thermal neutrality: forcing $a_1 = a_2 = 0$ costs 8.27 vs 3.66 train. Likewise $\text{corr}(\log \tau, \text{yield}) = +0.061$ is an interior optimum, not irrelevance.



Rate constants from the fitted parameters. Left of the crossover the reactor makes B faster than it destroys it; right of it, selectivity collapses.

Reaction	Activation energy
Desired $A \rightarrow B$	43.16 kJ/mol
Waste $B \rightarrow C$	250.07 kJ/mol

Why feed concentration does nothing ($r = +0.009$)	Adiabatic swing
$a_1 (A \rightarrow B) = -11.79 \text{ K} \cdot \text{L/mol}$	-27 K
$a_2 (B \rightarrow C) = +11.34 \text{ K} \cdot \text{L/mol}$	+26 K
Net	-1 K



We *recovered* the reactor, not a regressor

$$dCA/dz = -k_1 \cdot CA$$

$$dCB/dz = k_1 \cdot CA - k_2 \cdot CB$$

$$dT/dz = a_1 \cdot k_1 \cdot CA + a_2 \cdot k_2 \cdot CB + U \cdot (T_{\text{jacket}} - T)$$

7 physical parameters. Fixed functional form. No architecture search. The form came from the chemistry, so there was nothing to search.

Model	Protocol	RMSE
Best tree ensemble (+ physics features)	10f-CV	16.37
Physics ODE	10f-CV	6.36
Physics ODE	training fit	3.66

Three choices, each measured

- **Noise-averaged prediction** at $\sigma = 1.67$ K, fixed a priori from errors-in-variables, never tuned on the CV that judges it. Training worse (3.656 $\rightarrow$ 3.764), cross-validation better (6.358 $\rightarrow$ 5.928) on all three seeds: input noise, not overfitting.
- **A 13% ExtraTrees component**, weight derived not tuned: error correlation $\rho = 0.070$ gives a variance-optimal $w^* = 0.887$; we ship 0.87.
- **Validation**. Repeated 10-fold plus cold-start folds refit from scratch (5.98); two independent protocols agreeing.

A limit of our own protocol

Leave-one-seed-out underprices hyper-parameter selection: all seeds re-partition the same 150 rows. At row level that costs +0.371 RMSE, so 5.671 is a lower bound.

Why this approach for this problem

150 rows cannot support a flexible model at a sharp cliff; a tree cannot extrapolate past its outermost split. An ODE can, and an engineer can check its parameters against known kinetics.



What the score *actually* says

RMSE

11.0442

MAE

3.2408

 R^2

0.9061

Rows missed badly	Their error	Error on the other ~47
2	55.0	1.09
3	45.0	0.57

Independent replication of the physics

A separate blind refit: different integrator (LSODA), different optimizer, no access to our code or notes.

Parameter	Ours	Blind
E_1 (kJ/mol)	43.161	43.156
E_2 (kJ/mol)	250.072	250.53
a_1	-11.7919	-11.7961
train RMSE	3.6559	3.6554

Rank 13 of 400. The metrics disagree, and that is the finding: $RMSE / MAE = 3.41$ against 1.25 Gaussian, 1.41 Laplace. Too high for broadly mediocre prediction.

~47 of 50 rows predicted to about 1 yield-point. Two or three cliff rows missed by ~50. R^2 0.9061 implies a test spread of 36.0 vs 38.3 in training, and a two-sample classifier confirms the match (AUC 0.500).

What we got right, and what we did not

We pre-registered the mechanism: a few cliff rows carry nearly all squared error. The score confirms it, but we underestimated the magnitude. Our bootstrap put RMSE in [2.94, 8.10]; 11.04 sits outside. MAE we called: 3.24 observed, [1.74, 4.03] predicted. None of our 150 training rows missed by over 40 points; the test set produced two or three.



Where it fails, and how you would tell

If this started predicting badly tomorrow

Source	Symptom
Chemical	Fouling raises U · catalyst degradation shifts k_1 · feed change moves CA_0 · operation outside 351.6–548.0 K
Sensor / data	Near the cliff a 2 K temperature error becomes a ~20-point yield error. Median measured input offset 1.67 K ; 12 of 150 rows resist any offset within ± 25 K
ML / model	The cliff: 29% of rows carry 91% of squared error. Our blend also lifts 6 dead-reactor predictions above 5% yield (max 11.5)

The diagnostic, and it works only because the model is mechanistic

Refit the seven parameters and read which one moved

A black-box model tells you the prediction is wrong. It cannot tell you which piece of the plant changed.

What moves	Diagnosis
U alone	heat-transfer fouling
k_1 alone	catalyst degradation
CA_0 path only	feed composition change
Nothing moves, errors rise	sensor drift or unmodelled phenomenon

Known limits, stated plainly

150 rows, one reactor geometry. Tube diameter is constant across rows, so it folds into the fitted pre-exponentials; a different tube needs a refit, not a rescale. Cliff rows stay hard: 8 strategies tested after submission, none cleared our bar. Hence recovering the reactor rather than approximating it: not just a predictor, an instrument.